

Problem 1. Benchmarks, baselines and evaluation metrics for graph embedding on *homogeneous* graphs

A *homogeneous* graph $G = (V, E)$ carries a single node type and a single edge type: the type-mapping functions $\phi : V \rightarrow \mathcal{A}$ and $\psi : E \rightarrow \mathcal{R}$ satisfy $|\mathcal{A}| = |\mathcal{R}| = 1$. Optionally each node carries an attribute vector $x_v \in \mathbb{R}^d$ and a label y_v . Graph embedding learns a map $f : V \rightarrow \mathbb{R}^k$ with $k \ll |V|$ such that proximity in $\mathbb{R}^k$ reflects proximity in G ; the embeddings are then evaluated on

- **node classification** — predict y_v for held-out nodes, and
- **link prediction** — score unobserved pairs (u, v) so that true missing edges rank above non-edges.

The heterogeneous counterpart (multiple node/edge types, metapath-based methods) is the subject of Problem 2 and is deliberately not covered here.

Statistics in Table 1 were obtained by loading each graph and counting; the OGB figures in Table 2 are from the official OGB documentation. Where the literature reports conflicting values, the discrepancy is footnoted rather than silently resolved.

1. Benchmark datasets

1.1 Citation networks (the default benchmarks)

Cora, **CiteSeer**, **PubMed** (Sen et al., *Collective Classification in Network Data*, AI Magazine 29(3), 2008; split protocol from Yang, Cohen & Salakhutdinov, *Revisiting Semi-Supervised Learning with Graph Embeddings*, ICML 2016) are the three *Planetoid* citation graphs and remain the default testbed for semi-supervised node classification. Nodes are scientific publications, an undirected edge joins two papers when one cites the other, and the label is the paper’s topic.

Cora collects machine-learning papers and uses **binary bag-of-words** features.¹ **CiteSeer** is the sparsest of the three and has a number of isolated nodes, which is why absolute accuracies are lowest on it. **PubMed** collects diabetes-related biomedical abstracts and, unlike the other two, uses **real-valued TF-IDF** weights rather than binary indicators. All three are also standard *link prediction* benchmarks: the GAE/VGAE protocol hides 5% of edges for validation and 10% for test, samples an equal number of non-edges, and reports AUC and AP.

1.2 Co-authorship and co-purchase networks

Introduced by Shchur et al. (*Pitfalls of Graph Neural Network Evaluation*, NeurIPS 2018 Relational Representation Learning Workshop) precisely because results on the three Planetoid graphs with a single fixed split are not statistically reliable.

Coauthor CS and **Coauthor Physics** come from the KDD Cup 2016 Microsoft Academic Graph: an edge means two authors co-authored a paper, features are binary keyword indicators over an author’s papers, and the label is the author’s most active field of study. **Amazon Computers** and **Amazon Photo** are segments of the Amazon co-purchase graph: an edge means two products are frequently bought together, features are binary bag-of-words over product reviews, and the label is the product category. All four are used for node classification and, being attributed and undirected, also serve as link prediction benchmarks.

¹The Cora edge count is reported inconsistently: 5,429 is the number of lines in the original `cora.cites` file (which contains duplicate and self-referential entries), 5,278 is the number of distinct undirected edges after de-duplication, and 10,556 is the number of *directed* entries stored when each undirected edge is materialised in both directions. All three describe the same graph.

1.3 Web-page and heterophilous benchmarks (the counterexamples)

The benchmarks above are strongly *homophilous*, and a GNN that simply averages over neighbourhoods does well on them almost by construction. The suite assembled by Pei et al. (*Geom-GCN*, ICLR 2020) and popularised by Zhu et al. (*Beyond Homophily in Graph Neural Networks*, NeurIPS 2020) exists to break that assumption.

WebKB (Cornell, Texas, Wisconsin) are university web pages linked by hyperlinks, with binary bag-of-words features and 5 page classes (*student*, *project*, *course*, *staff*, *faculty*); they are very small and very heterophilous, so variance across splits is large and results must be averaged over the 10 standard splits. **Actor** is the actor-only induced subgraph of a film–director–actor–writer network, where an edge means two actors are co-mentioned on the same Wikipedia page. **Chameleon** and **Squirrel** are Wikipedia page–page networks on those topics, with nodes binned into 5 classes by average monthly traffic. All six are used almost exclusively for node classification. The quantity that separates the two groups is the **edge homophily ratio**

$$h_{\text{edge}}(G) = \frac{|\{(u, v) \in E : y_u = y_v\}|}{|E|}, \quad (1)$$

which we measured directly on each graph. The values appear in the last column of Table 1 and range from 0.93 (Coauthor Physics) down to 0.11 (Texas); Figure 2 illustrates the two regimes.

The edge set used must be stated, because two choices in it move these numbers.² The ordering of the datasets is unaffected either way: the six web/Wikipedia/co-occurrence graphs sit far below the citation, co-authorship and co-purchase graphs on any of these definitions, which is the only claim the rest of this answer relies on.

1.4 Social and interaction networks

These are the benchmarks of the *unsupervised* embedding literature (DeepWalk, LINE, node2vec). They are typically **feature-less** and **multi-label**, which is why Micro-F1/Macro-F1 rather than accuracy became the standard metric in that line of work.

BlogCatalog is a friendship network of bloggers labelled with overlapping interests and **Flickr** is a contact network labelled with the groups each user joined; both are from Tang & Liu, *Relational Learning via Latent Social Dimensions*, KDD 2009. **YouTube** is a user–user social network labelled with group memberships, from the companion paper Tang & Liu, *Scalable Learning of Collective Behavior based on Sparse Social Dimensions*, CIKM 2009, and is the scalability stress test of that literature. (Statistics for all three are as tabulated in the DeepWalk paper.) None has node attributes, so structure is the only signal. **Reddit** (Hamilton, Ying & Leskovec, NIPS 2017) links

²(i) *Which edge set.* We evaluate Eq. (1) over *distinct unordered pairs* — precisely the set counted by the $|E|$ column of Table 1, so the ratio and its denominator refer to the same object. The six directed graphs also admit a count over the stored directed entries, which weights a reciprocated arc twice as heavily as a one-way arc; that gives slightly different values (e.g. Chameleon 0.235 rather than 0.231) and agrees less well with the literature. (ii) *Self-loops.* Texas and Wisconsin carry 16 each, Actor 93 and Squirrel 140, and a self-loop trivially satisfies $y_u = y_v$, so it always counts as homophilous. Following Zhu et al. we include them; excluding them would give Texas 0.06 rather than 0.11 and Wisconsin 0.18 rather than 0.21 — a large relative change on exactly the datasets whose heterophily is the point. Under the stated convention our values reproduce Zhu et al.’s published figures *exactly* to two decimal places for Cora, CiteSeer, PubMed, Texas, Wisconsin, Actor, Chameleon and Squirrel. **Cornell is the sole exception:** Zhu et al. report 0.30 and we measure 0.13. We could not reproduce 0.30 under any definition we tried — over unordered pairs 0.132, the same excluding self-loops 0.123, over the stored directed entries 0.131, or node homophily in the sense of Pei et al. 0.186 — so we report the measured value and flag the disagreement rather than repeating a figure we cannot reproduce.

two posts when the same user commented on both, labels each post with its community, and is the standard *inductive, large-scale* node-classification benchmark.³

Warning — three names, six graphs. *BlogCatalog*, *Flickr* and *PPI* each denote two genuinely different graphs in this literature, and numbers are frequently mixed up between them. The feature-less multi-label versions above are the ones used by DeepWalk/node2vec; the attributed versions (BlogCatalog with 5,196 nodes / 171,743 edges / 8,189 features / 6 classes; the GraphSAINT Flickr with 89,250 nodes / 449,878 edges / 500 features / 7 classes) are the ones shipped by modern GNN libraries. A paper’s numbers are only comparable to another’s if the same version is used.

1.5 Biological networks

PPI (protein–protein interaction, GraphSAGE version) is **24 separate graphs**, one per human tissue, whose node features are positional gene sets, motif gene sets and immunological signatures, and whose **121 binary labels** are gene ontology terms. The split is *inductive across graphs* — 20 train, 2 validate, 2 test — so the test graphs are never seen during training. This is the canonical test of whether a method generalises to unseen graphs at all, which transductive methods such as DeepWalk cannot do by construction. The **node2vec PPI** is a different graph: a single *Homo sapiens* subgraph with 3,890 nodes, 76,584 edges and 50 labels.

1.6 The Open Graph Benchmark (OGB)

Hu et al. (*Open Graph Benchmark: Datasets for Machine Learning on Graphs*, NeurIPS 2020) introduced OGB to fix the two standing problems with the classic benchmarks: they are tiny, and their random splits are unrealistically easy. OGB fixes the split, fixes the metric, and provides a shared evaluator, so numbers are comparable across papers. Statistics are in Table 2. The homogeneous members are:

- **ogbn-arxiv** — directed citation graph of CS arXiv papers, node features the averaged word2vec embedding of title and abstract, labels the 40 subject areas. *Time split*: train up to 2017, validate on 2018, test from 2019.
- **ogbn-products** — Amazon co-purchase graph, bag-of-words-plus-PCA features, 47 top-level categories. *Sales-rank split*: the most popular products train and the rest are validation/test, which is much harder than a random split.
- **ogbn-papers100M** — the same construction as ogbn-arxiv at 111M nodes; only ~ 1.5 M are labelled (172 subject areas). The standard scalability benchmark.
- **ogbn-proteins** — proteins from 8 species joined by weighted association edges. It carries *edge* features and *no* node features, and its 112 binary function-prediction tasks are split *by species* and scored by average ROC-AUC.
- **ogbl-collab** (author collaboration), **ogbl-ppa** (protein association), **ogbl-citation2** (paper citation) and **ogbl-ddi** (drug–drug interaction) are the four homogeneous *link prediction* tasks. Each fixes its own split, metric and negative set in Table 2; ogbl-ddi is notable for having *no* node features at all, so only structure can be used.

³We measured 57,307,946 distinct undirected edges (114,615,892 directed entries). This reproduces the average degree of 492 stated in the GraphSAGE paper ($2 \times 57,307,946 / 232,965 = 492.0$); the figure of 11.6M edges that appears on some dataset pages refers to a sparsified copy of the graph.

1.7 Split protocols — and why they must be stated

A result is meaningless without its split, because the same model can differ by many accuracy points across protocols on the same graph:

- **Planetoid semi-supervised split:** exactly **20 labelled nodes per class** for training, 500 for validation, 1,000 for test. This is the split on which the familiar Cora $\approx 81.5\%$ figure for GCN is reported.
- **Full-supervised / “Geom-GCN” split:** **60/20/20%** of nodes per class, averaged over 10 random splits. Standard for the heterophilous benchmarks and typically 5–10 points higher than the 20-per-class numbers.
- **Random splits with fixed label budget** (Shchur et al.): the Planetoid budget but re-randomised, reported as mean $\pm$ std over ≥ 100 splits. This was the paper’s central recommendation, since the single fixed Planetoid split does not support the ranking of methods usually claimed from it.
- **OGB splits** are fixed and non-random — by time, by sales rank, by species, by experimental throughput, by protein target — and deliberately induce distribution shift between train and test.
- **Link prediction:** edges, not nodes, are split. The message-passing graph must contain *only* training edges, otherwise the held-out edge leaks into its own prediction. See Figure 3.

2. Graph schemas of the benchmarks

The *graph schema* is the meta-template of a network: the graph over node types whose edges are the permitted relation types. For a homogeneous graph the schema is by definition **degenerate** — one node type and one edge type — so it is a single vertex with a single self-loop:

$$\mathcal{T}_G = (\{A\}, \{A \xrightarrow{R} A\}), \quad |\mathcal{A}| = |\mathcal{R}| = 1.$$

This is the honest answer to “draw the related graph schema” for these benchmarks, and it is exactly what distinguishes them from the heterogeneous datasets of Problem 2, whose schemas have several node types and several relation types. What actually differs between the homogeneous benchmarks is the *semantics* instantiated into A and R , so the benchmarks are grouped into seven schema families in Figure 1. Because a one-node schema is visually vacuous on its own, Figure 2 additionally sketches the local structure that the schema does *not* capture and which nevertheless drives model performance: the label homophily of Eq. (1).

3. Baselines

3.1 Matrix factorisation and spectral methods

- **Laplacian Eigenmaps** — Belkin & Niyogi, *Laplacian Eigenmaps and Spectral Techniques for Embedding and Clustering*, NIPS 2001; expanded as *Laplacian Eigenmaps for Dimensionality Reduction and Data Representation*, *Neural Computation* 15(6):1373–1396, 2003 (the canonical citation). Embeds nodes by the eigenvectors of the graph Laplacian $L = D - A$ belonging to the smallest non-zero eigenvalues, i.e. it minimises $\sum_{(u,v) \in E} w_{uv} \|f(u) - f(v)\|^2$ subject to an orthogonality constraint ruling out the trivial solution. The ancestor of essentially every proximity-preserving embedding; spectral clustering is the same computation.
- **GraRep** — Cao, Lu & Xu, *GraRep: Learning Graph Representations with Global Structural Information*, CIKM 2015. Observes that first-order proximity is not enough, and factorises

Table 1: Homogeneous node-classification / link-prediction benchmarks. **dir.:** **U** = the stored edge set is symmetric (undirected), **D** = it is not (directed). $|E|$ counts *distinct unordered pairs*, with self-loops counted once; the parenthesised figure is the number of entries in the stored edge list, which is $2|E|$ for the U rows but not for the D rows, since a directed graph stores only the arcs that exist. h_{edge} is Eq. (1) measured on the graph, its denominator being exactly the $|E|$ of the preceding column — the same distinct unordered pairs, self-loops included (see the footnote; this convention matters, and the alternatives are listed there). “bin.” = binary bag-of-words, “TF-IDF” / “real” = real-valued, ML = multi-label. [†]Cornell: see footnote 2.

Family	Dataset	$ V $	$ E $	d	#cls	feat.	dir.	h_{edge}
Citation	Cora	2,708	5,278 (10,556)	1,433	7	bin.	U	0.81
	CiteSeer	3,327	4,552 (9,104)	3,703	6	bin.	U	0.74
	PubMed	19,717	44,324 (88,648)	500	3	TF-IDF	U	0.80
Co-author	Coauthor CS	18,333	81,894	6,805	15	bin.	U	0.81
	Coauthor Physics	34,493	247,962	8,415	5	bin.	U	0.93
Co-purchase	Amazon Computers	13,752	245,861	767	10	bin.	U	0.78
	Amazon Photo	7,650	119,081	745	8	bin.	U	0.83
Web / hyperlink	Cornell	183	280 (298)	1,703	5	bin.	D	0.13 [†]
	Texas	183	295 (325)	1,703	5	bin.	D	0.11
	Wisconsin	251	466 (515)	1,703	5	bin.	D	0.21
	Chameleon	2,277	31,421 (36,101)	2,325	5	bin.	D	0.23
	Squirrel	5,201	198,493 (217,073)	2,089	5	bin.	D	0.22
Co-occurrence	Actor	7,600	26,752 (30,019)	932	5	bin.	D	0.22
Social	BlogCatalog	10,312	333,983	—	39	none (ML)	U	—
	Flickr	80,513	5,899,882	—	195	none (ML)	U	—
	YouTube	1,138,499	2,990,443	—	47	none (ML)	U	—
	Reddit	232,965	57,307,946	602	41	real	U	—
Biological	PPI (24 graphs)	56,944	793,632	50	121	real (ML)	U	—

Table 2: The homogeneous Open Graph Benchmark datasets. Edge counts are as published by OGB (undirected counts for undirected graphs).

Dataset	Domain	$ V $	$ E $	Node features	Split	Metric
<i>Node classification</i>						
ogbn-arxiv	citation	169,343	1,166,243	128 word2vec	time	Accuracy
ogbn-products	co-purchase	2,449,029	61,859,140	100 BoW+PCA	sales rank	Accuracy
ogbn-papers100M	citation	111,059,956	1,615,685,872	128 word2vec	time	Accuracy
ogbn-proteins	protein	132,534	39,561,252	none (8-d edges)	species	ROC-AUC
<i>Link prediction</i>						
ogbl-collab	co-authorship	235,868	1,285,465	128 word2vec	time	Hits@50
ogbl-ppa	protein	576,289	30,326,273	58-d one-hot	throughput	Hits@100
ogbl-citation2	citation	2,927,963	30,561,187	128 word2vec	time	MRR
ogbl-ddi	drug-drug	4,267	1,334,889	none	protein target	Hits@20

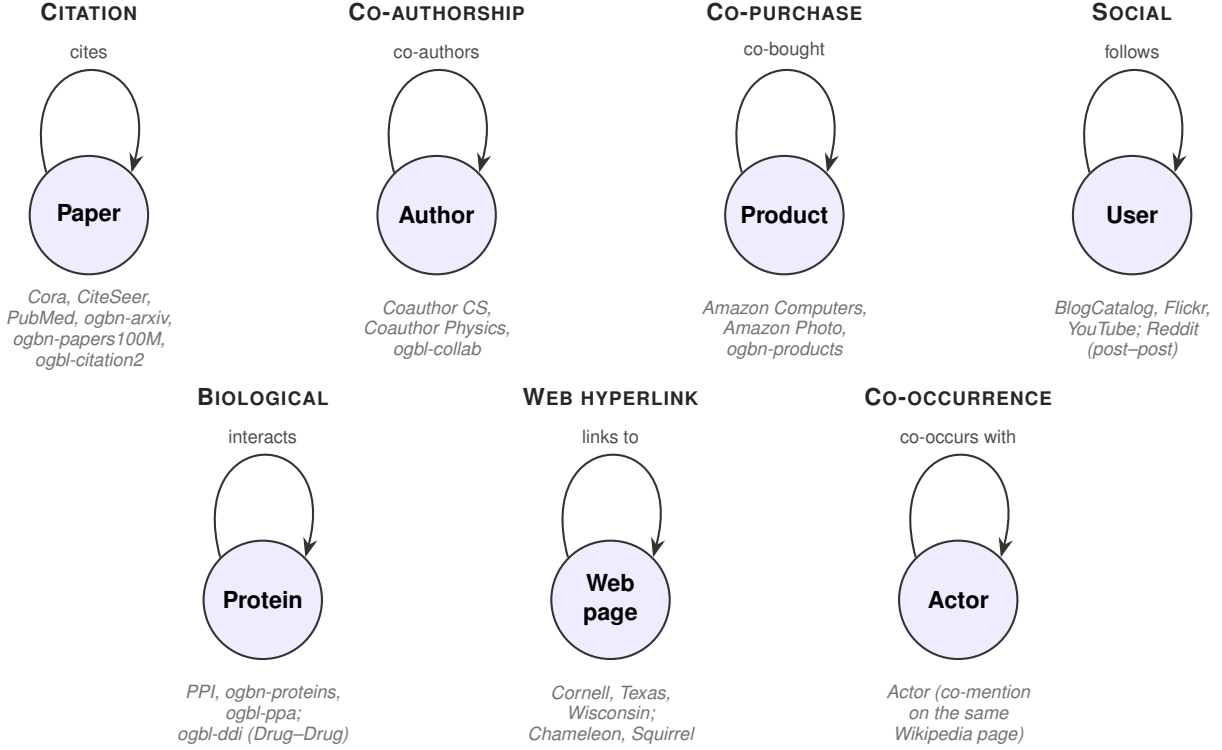


Figure 1: Graph schemas of the homogeneous benchmarks, grouped into seven families. Every schema is a single node type with a single self-relation; only the interpretation of the node and the edge changes between families.

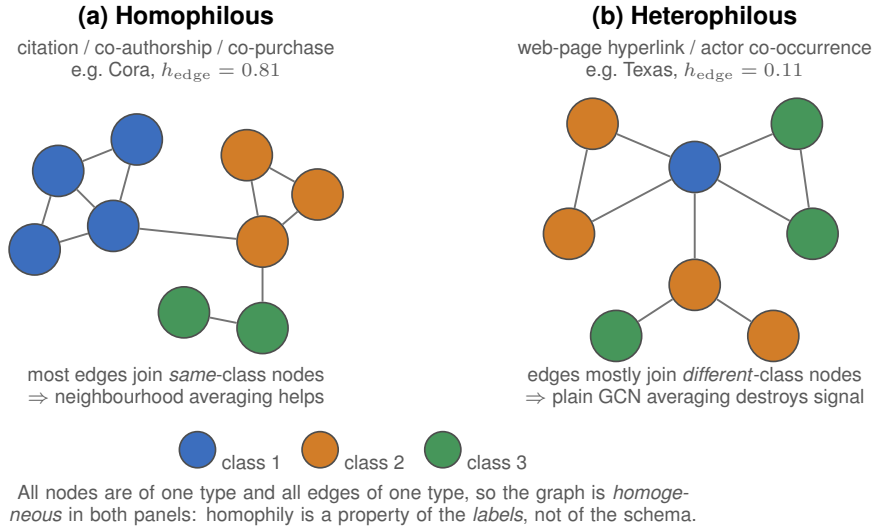


Figure 2: Illustrative subgraphs. Both graphs have the same (degenerate) schema; they differ in how labels are distributed over edges. The measured h_{edge} values are those of Table 1.

the k -step transition matrices $P^1, \dots, P^K$ (via SVD of their log-transformed positive entries) separately, concatenating the K resulting embeddings. This captures k -hop structure explicitly, at the cost of dense $|V| \times |V|$ powers.

- **HOPE** — Ou, Cui, Pei, Zhang & Zhu, *Asymmetric Transitivity Preserving Graph Embedding*, KDD 2016. Targets *directed* graphs, where proximity is asymmetric. It learns two vectors per node (source and target) and factorises a high-order proximity matrix — Katz, Rooted PageRank, Common Neighbors or Adamic–Adar — which it shows can be written in a sparse form $S = M_g^{-1} M_\ell$ admitting a generalised SVD, so the cost is linear in $|E|$.
- **NetMF** — Qiu, Dong, Ma, Li, Wang & Tang, *Network Embedding as Matrix Factorization: Unifying DeepWalk, LINE, PTE, and node2vec*, WSDM 2018. Proves that the skip-gram-with-negative-sampling objectives of DeepWalk, LINE, PTE and node2vec all implicitly factorise a closed-form matrix, for DeepWalk $\log(\frac{\text{vol}(G)}{bT} (\sum_{r=1}^T P^r) D^{-1})$. Factorising that matrix directly gives better embeddings than running the sampler, and the paper is the standard theoretical bridge between the random-walk and factorisation families.

3.2 Random-walk / skip-gram methods

- **DeepWalk** — Perozzi, Al-Rfou & Skiena, *DeepWalk: Online Learning of Social Representations*, KDD 2014. The paper that started the field. It samples uniform random walks from each node, treats each walk as a “sentence” and each node as a “word”, and runs word2vec/skip-gram with hierarchical softmax over them. Nodes appearing in similar walk contexts receive similar vectors. It requires no node attributes and is trivially parallel, but is transductive: unseen nodes have no embedding.
- **LINE** — Tang, Qu, Wang, Zhang, Yan & Mei, *LINE: Large-scale Information Network Embedding*, WWW 2015. Dispenses with walks and optimises two explicit objectives: *first-order* proximity (observed edges, $p_1(u, v) = \sigma(f(u)^\top f(v))$) and *second-order* proximity (shared neighbourhoods, a context embedding per node), each trained by negative sampling with edge-sampling to handle weights. The two embeddings are concatenated. It scales to millions of nodes and is the standard “no-walk” baseline.
- **node2vec** — Grover & Leskovec, *node2vec: Scalable Feature Learning for Networks*, KDD 2016. Generalises DeepWalk by biasing the walk with return parameter p and in-out parameter q , interpolating between breadth-first search (which captures *structural equivalence*) and depth-first search (which captures *homophily*). Tuning (p, q) per dataset is what buys its consistent gain over DeepWalk. It also defines the standard Hadamard/L1/L2 binary operators for turning node embeddings into edge features for link prediction.
- **struc2vec** — Ribeiro, Savarese & Figueiredo, *struc2vec: Learning Node Representations from Structural Identity*, KDD 2017. Argues that nodes with similar *roles* should embed together even if they are far apart in the graph — something no proximity-based method can do. It builds a multilayer context graph whose layer- k edge weights compare the ordered degree sequences of k -hop neighbourhoods, then runs skip-gram on walks over that auxiliary graph.

3.3 Autoencoder methods

- **SDNE** — Wang, Cui & Zhu, *Structural Deep Network Embedding*, KDD 2016. A deep autoencoder whose input is a node’s adjacency row. It combines a reconstruction loss on that row

(second-order proximity), re-weighted so that the many zeros do not dominate the few ones, with a Laplacian-eigenmaps-style penalty tying connected nodes together (first-order proximity). The first non-linear (deep) network embedding method.

- **GAE / VGAE** — Kipf & Welling, *Variational Graph Auto-Encoders*, NIPS 2016 Bayesian Deep Learning Workshop. Uses a two-layer GCN as the encoder, $Z = \text{GCN}(X, A)$, and a simple inner-product decoder $\hat{A} = \sigma(ZZ^\top)$, trained by reconstruction cross-entropy; the variational version places a Gaussian prior on Z and adds a KL term. It is the default *unsupervised* baseline for link prediction on Cora/CiteSeer/PubMed.

3.4 Graph neural networks (message passing)

All of these compute $h_v^{(l+1)} = \text{UPDATE}(h_v^{(l)}, \text{AGGREGATE}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$ and differ in the two functions.

- **GCN** — Kipf & Welling, *Semi-Supervised Classification with Graph Convolutional Networks*, ICLR 2017. A first-order approximation of a Chebyshev spectral filter, giving the layer $H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$ with $\tilde{A} = A + I$. It is a symmetrically-normalised neighbourhood average followed by a linear map; two layers is the standard configuration. The single most reported baseline in the field.
- **GraphSAGE** — Hamilton, Ying & Leskovec, *Inductive Representation Learning on Large Graphs*, NIPS 2017. Makes GNNs *inductive* and scalable: instead of using the full adjacency it samples a fixed number of neighbours per hop and learns an aggregator (mean, LSTM or max-pool) plus a concatenation with the node’s own previous representation. Because the aggregator is shared, embeddings can be produced for nodes — and graphs — never seen in training. Introduced the Reddit and PPI benchmarks.
- **GAT** — Veličković, Cucurull, Casanova, Romero, Liò & Bengio, *Graph Attention Networks*, ICLR 2018. Replaces the fixed normalisation coefficient $1/\sqrt{d_u d_v}$ with a learned attention weight $\alpha_{uv} = \text{softmax}_u(\text{LeakyReLU}(a^\top [W h_v \| W h_u]))$, so a node can weight its neighbours unequally. Multi-head attention stabilises training. No spectral decomposition and no knowledge of the full graph is required, so it too is inductive.
- **JK-Net** — Xu, Li, Tian, Sonobe, Kawarabayashi & Jegelka, *Representation Learning on Graphs with Jumping Knowledge Networks*, ICML 2018. Observes that the right neighbourhood radius differs per node (hub vs. periphery), and lets the final representation “jump” by combining *all* intermediate layers $h^{(1)}, \dots, h^{(L)}$ via concatenation, max-pooling or an LSTM attention, rather than using only the last. A standard remedy for over-smoothing.
- **GIN** — Xu, Hu, Leskovec & Jegelka, *How Powerful are Graph Neural Networks?*, ICLR 2019. Analyses message passing through the Weisfeiler–Lehman test and shows that mean and max aggregation lose information; a *sum* aggregator with an MLP update, $h_v^{(l+1)} = \text{MLP}((1 + \epsilon)h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l)})$, is maximally expressive, i.e. as strong as 1-WL. The reference point for all expressive-power arguments.
- **SGC** — Wu, Souza, Zhang, Fifty, Yu & Weinberger, *Simplifying Graph Convolutional Networks*, ICML 2019. Removes every non-linearity from a K -layer GCN, collapsing it to $\hat{Y} = \text{softmax}(S^K X W)$ with $S = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$. Since $S^K X$ is a fixed pre-computation, training is

plain logistic regression — orders of magnitude faster, and on the homophilous citation benchmarks essentially as accurate. An important sanity baseline: it measures how much the non-linearity was really contributing.

- **APPNP** — Gasteiger (formerly Klicpera), Bojchevski & Günnemann, *Predict then Propagate: Graph Neural Networks meet Personalized PageRank*, ICLR 2019. Decouples prediction from propagation: an MLP produces per-node predictions H , which are then diffused by a truncated personalized-PageRank iteration $Z^{(k+1)} = (1 - \alpha)\hat{A}Z^{(k)} + \alpha H$. The teleport term α keeps the node’s own signal alive, so a large propagation radius is possible without over-smoothing and without extra parameters.
- **GCNII** — Chen, Wei, Huang, Ding & Li, *Simple and Deep Graph Convolutional Networks*, ICML 2020. Makes genuinely deep GCNs (32–64 layers) work using two ingredients: an *initial residual* connection to $H^{(0)}$ (the APPNP teleport idea) and *identity mapping* of the weight matrix, $(1 - \beta_l)I + \beta_l W^{(l)}$. Proves the result still expresses a polynomial filter of arbitrary order and so does not collapse to over-smoothing.
- **SEAL** — Zhang & Chen, *Link Prediction Based on Graph Neural Networks*, NeurIPS 2018. Reframes link prediction as *subgraph classification*: for a candidate pair (u, v) it extracts the h -hop enclosing subgraph, applies a Double-Radius Node Labeling that encodes each node’s distance to both u and v , and classifies the labelled subgraph with a GNN. The γ -decaying theory in the paper shows such a GNN can learn any of the classical heuristics below rather than assuming one. The strongest general-purpose link-prediction baseline.
- **NBFNet** — Zhu, Zhang, Xhonneux & Tang, *Neural Bellman-Ford Networks: A General Graph Neural Network Framework for Link Prediction*, NeurIPS 2021. Generalises the Bellman–Ford path formulation by replacing $(\min, +)$ with learned INDICATOR, MESSAGE and AGGREGATE operators, so the score for (u, v) is a learned sum over all paths from u to v . Inductive (it depends on structure relative to u , not on node identity) and applies to homogeneous graphs and knowledge graphs alike.

3.5 Self-supervised and contrastive methods

- **DGI** — Veličković, Fedus, Hamilton, Liò, Bengio & Hjelm, *Deep Graph Infomax*, ICLR 2019. Maximises the mutual information between patch representations h_v and a global summary s of the graph, using a discriminator trained to tell real (h_v, s) pairs from corrupted ones (the corruption shuffles the node features). Needs no random walks and no labels, and was the first unsupervised method to match supervised GCN on the citation benchmarks.
- **GRACE** — Zhu, Xu, Yu, Liu, Wu & Wang, *Deep Graph Contrastive Representation Learning*, ICML 2020 Workshop on Graph Representation Learning and Beyond. A node-level SimCLR analogue: generate two views by removing edges and masking features, then maximise agreement between the two representations of the same node against all other nodes as negatives (InfoNCE). Contrasts at node rather than graph-summary level, which DGI does not.
- **GraphCL** — You, Chen, Sui, Chen, Wang & Shen, *Graph Contrastive Learning with Augmentations*, NeurIPS 2020. Systematically studies four graph augmentations — node dropping, edge perturbation, attribute masking and subgraph sampling — and shows which combinations help on which domain. Mostly graph-level, but its augmentation study is the reference for the whole contrastive family.

- **BGRL** — Thakoor, Tallec, Azar, Azabou, Dyer, Munos, Veličković & Valko, *Large-Scale Representation Learning on Graphs via Bootstrapping*, ICLR 2022. Removes negative samples altogether using a BYOL-style online/target encoder pair with a stop-gradient, cutting memory from quadratic to linear in the number of nodes. This is what makes self-supervised learning feasible at OGB scale.

3.6 Non-learned link-prediction heuristics

Every link-prediction paper reports these; they are strong, free, and a learned method that does not beat them is not worth its compute. The framing is due to Liben-Nowell & Kleinberg (*The Link Prediction Problem for Social Networks*, CIKM 2003; journal version JASIST 58(7), 2007). Let $\Gamma(u)$ be the neighbour set of u and A the adjacency matrix.

Heuristic	Score $s(u, v)$	Reference / note
Common Neighbors	$ \Gamma(u) \cap \Gamma(v) $	Liben-Nowell & Kleinberg, CIKM 2003
Jaccard	$ \Gamma(u) \cap \Gamma(v) / \Gamma(u) \cup \Gamma(v) $	normalises CN by degree
Adamic-Adar	$\sum_{w \in \Gamma(u) \cap \Gamma(v)} 1 / \log \Gamma(w) $	Adamic & Adar, <i>Friends and Neighbors on the Web</i> , Social Networks 25(3), 2003
Resource Allocation	$\sum_{w \in \Gamma(u) \cap \Gamma(v)} 1 / \Gamma(w) $	Zhou, Lü & Zhang, <i>Eur. Phys. J. B</i> 71(4), 2009; penalises hubs harder than AA
Preferential Attachment	$ \Gamma(u) \cdot \Gamma(v) $	Barabási & Albert, <i>Science</i> 286, 1999
Katz	$\sum_{\ell \geq 1} \beta^\ell \text{paths}_{u,v}^{(\ell)} = ((I - \beta A)^{-1} - I)_{uv}$	Katz, <i>Psychometrika</i> 18(1), 1953; global, needs $\beta < 1/\lambda_{\max}$
Personalized PageRank	stationary probability at v of a random walk from u with restart	Jeh & Widom, <i>Scaling Personalized Web Search</i> , WWW 2003; PageRank itself: Page, Brin, Motwani & Winograd, Stanford InfoLab TR, 1999
SimRank	$\frac{\gamma}{ \Gamma(u) \Gamma(v) } \sum_{a \in \Gamma(u)} \sum_{b \in \Gamma(v)} s(a, b)$	Jeh & Widom, KDD 2002; “similar if referenced by similar nodes”

The first five are *local* (2-hop) and therefore cheap but blind to long-range structure; Katz, PPR and SimRank are *global* and stronger but cost a matrix inverse or an iterative solve. On ogbl-ddi and ogbl-ppa, Common Neighbors and Adamic-Adar remain competitive with many GNNs, which is the usual argument for reporting them.

4. Evaluation metrics

4.1 Node classification

Let C be the number of classes and $\text{TP}_c, \text{FP}_c, \text{FN}_c$ the per-class counts.

- **Accuracy** = $\frac{1}{|V_{\text{test}}|} \sum_v \mathbf{1}[\hat{y}_v = y_v]$. The default for single-label benchmarks (Planetoid, OGB node tasks). Misleading under class imbalance.
- **Precision / Recall / F1** per class: $P_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}$, $R_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}$, $F1_c = \frac{2P_c R_c}{P_c + R_c}$.
- **Micro-F1** — pool the confusion matrix over all classes first, then compute one F1:

$$\text{Micro-F1} = \frac{2 \sum_c \text{TP}_c}{2 \sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c}. \quad (2)$$

In single-label multi-class classification every error is simultaneously one FP and one FN, so **Micro-F1 = Accuracy**. It is dominated by the frequent classes.

- **Macro-F1** — the *unweighted* mean of the per-class F1 scores:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (3)$$

Because every class contributes equally regardless of size, a model that ignores a rare class is punished. **This is why Macro-F1 matters:** on BlogCatalog or Flickr, where label frequencies span orders of magnitude, a model can score well on Micro-F1 while never predicting most of the 39 (resp. 195) labels — the gap between the two numbers is the diagnostic, and reporting only Micro-F1 hides it.

- **ROC-AUC**, computed one-vs-rest and averaged over classes for multi-class problems, and averaged over tasks for multi-label problems (this is the ogbn-proteins metric). It is threshold-free, so it measures ranking quality rather than the quality of a particular decision threshold.
- **Reporting convention.** A single number is not a result. The convention is **mean $\pm$ standard deviation over 10 (OGB, fixed split, 10 seeds) or 100 (Shchur et al., random splits) runs**, since the spread across splits on the small benchmarks routinely exceeds the differences between published methods.
- **Protocol for unsupervised embeddings.** DeepWalk/LINE/node2vec/DGI are evaluated by the *linear-probe* protocol: learn the embedding with no labels, **freeze** it, train a one-vs-rest ℓ_2 -regularised logistic regression on the embeddings of a fraction ρ of nodes, and report Micro-F1 and Macro-F1 on the rest. Results are given as a curve over training ratios $\rho \in \{10\%, \dots, 90\%\}$ (BlogCatalog) or $\{1\%, \dots, 10\%\}$ (Flickr, YouTube). Freezing is essential — fine-tuning the encoder would measure something else.

4.2 Link prediction

Link prediction is extremely class-imbalanced (a graph with $|V|$ nodes has $\binom{|V|}{2} - |E|$ non-edges), so the metric choice matters far more than for node classification. Positives are held-out edges; negatives are sampled node pairs that are not edges (Figure 3).

- **AUC-ROC** — the probability that a random positive outranks a random negative. Standard in the GAE/VGAE protocol with a 1:1 positive:negative ratio.
- **Average Precision (AP / AUPRC)** —

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k, \quad (4)$$

with P_k, R_k the precision and recall at threshold rank k . Unlike AUC it does not credit a model for correctly ranking the vast easy majority of negatives.

- **Precision@K / Recall@K** — the fraction of the top- K predicted pairs that are true edges, and the fraction of true edges recovered in the top K .

- **Hits@K** — the fraction of positive edges ranked in the top K against the sampled negatives:

$$\text{Hits@K} = \frac{1}{|E_{\text{test}}|} \sum_{e \in E_{\text{test}}} \mathbf{1}[\text{rank}(e) \leq K]. \quad (5)$$

OGB fixes K per dataset so that numbers are comparable: **Hits@50** for ogbl-collab (against 100,000 shared negatives), **Hits@20** for ogbl-ddi, **Hits@100** for ogbl-ppa (against 3,000,000 negatives).

- **MRR** — for each query (in ogbl-citation2, each source paper, ranked against 1,000 sampled negative references),

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}, \quad (6)$$

where rank_q is the position of the first true item for query q . Rewards getting the answer at the very top far more than at position 10.

- **NDCG@K** — with rel_i the relevance of the item at rank i ,

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \quad \text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}, \quad (7)$$

IDCG@K being the DCG of the ideal ordering. Used when a node has many true links of differing importance (recommendation-style evaluation).

Negative sampling, and why AUC is optimistic. The negatives define the metric. Three conventions are in use: (i) 1:1 *random negatives* (the GAE protocol), (ii) *a shared fixed negative set* for all positives (ogbl-collab, ogbl-ddi, ogbl-ppa), and (iii) *per-query negatives* (ogbl-citation2: 1,000 previously-published papers per source node). Real graphs are sparse, so a uniformly sampled pair is almost always a pair of distant, low-degree nodes that any model rejects trivially. AUC averages over all such pairs and therefore saturates near 1.0 and stops discriminating between methods — this is precisely why OGB abandoned it in favour of Hits@K and MRR, which only reward ranking the positive above the *hardest* sampled negatives. AP is likewise more informative than AUC because it weights the high-precision region where the model actually operates. A corollary is that Hits@K numbers are *only* comparable when the negative set is identical, which is the reason OGB ships a fixed evaluator.

4.3 Which metric is reported on which dataset

5. Summary

For homogeneous graphs the schema is degenerate — one node type, one edge type (Figure 1) — so what distinguishes the benchmarks is scale, feature type, and above all label homophily (Eq. (1)). The suite has moved in three waves: the small homophilous citation graphs with a fixed 20-per-class split, the re-splittable co-authorship/co-purchase graphs and the heterophilous WebKB/Wikipedia graphs introduced to expose the homophily assumption, and finally OGB, with realistic non-random splits and a shared evaluator at a scale where sampling-based training is mandatory. On metrics, the two points that matter most are that Micro-F1 equals accuracy in the single-label case while Macro-F1 is what exposes neglected rare classes, and that on sparse graphs AUC is optimistic, so ranking metrics against a *fixed* negative set (Hits@K, MRR) are what make link-prediction results comparable.

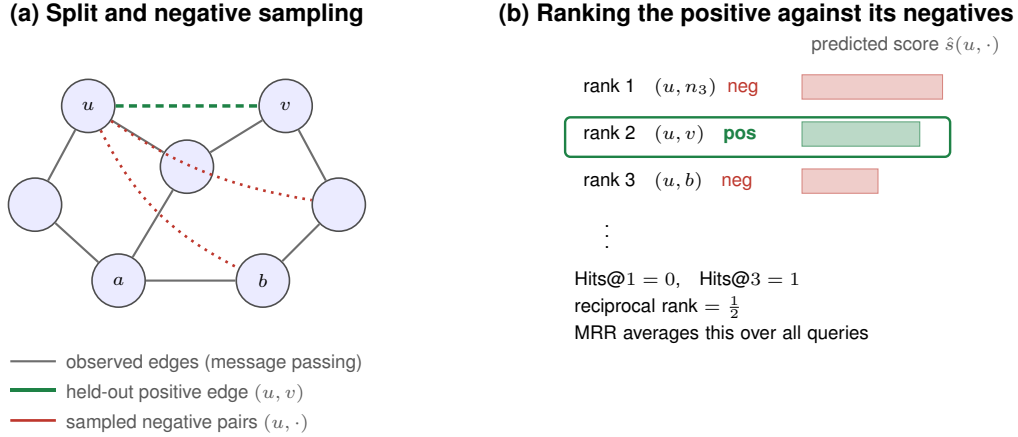


Figure 3: The link-prediction evaluation protocol. The held-out positive edge must be removed from the message-passing graph, and is scored against a sampled set of negatives; Hits@K and MRR are read off the resulting ranking.

Table 3: Dataset $\rightarrow$ task $\rightarrow$ metric normally reported.

Dataset	Task	Metric normally reported
Cora / CiteSeer / PubMed	node classification	Accuracy (= Micro-F1), mean $\pm$ std over 10–100 splits
Cora / CiteSeer / PubMed	link prediction	AUC-ROC and AP (GAE protocol, 85/5/10 edge split)
Coauthor CS / Physics	node classification	Accuracy, mean $\pm$ std over 100 random splits
Amazon Computers / Photo	node classification	Accuracy, mean $\pm$ std over 100 random splits
WebKB, Actor, Chameleon, Squirrel	node classification	Accuracy over the 10 fixed 60/20/20 splits
BlogCatalog / Flickr / YouTube	node classification, multi-label	Micro-F1 and Macro-F1 vs. training ratio
PPI (24 graphs)	inductive node classification, multi-label	Micro-F1
Reddit	inductive node classification	Micro-F1 (= Accuracy)
ogbn-arxiv / products / papers100M	node classification	Accuracy (fixed OGB split)
ogbn-proteins	node classification, 112 tasks	ROC-AUC averaged over tasks
ogbl-collab	link prediction	Hits@50 (100,000 shared negatives)
ogbl-ddi	link prediction	Hits@20 ($\sim$ 100,000 shared negatives)
ogbl-ppa	link prediction	Hits@100 (3,000,000 shared negatives)
ogbl-citation2	link prediction	MRR (1,000 negatives per source node)